Chapter 5
Orthogonality

5.1 The Scalar Product in $\mathbb{R}^n$

Overview:

- In the 4 previous chapters, we studied vector spaces and linear transformations.
  - Structures: vector addition, scalar multiplication
  - We can form linear combinations, solve equations etc
- However, some very basic concepts are missing:
  - How long is a vector?
  - What's the ***distance*** between 2 vectors?
  - What's the ***angle*** between 2 vectors? When are they ***perpendicular***垂直?
- We can answer all these questions by defining a **scalar product** on the vector space.

Scalar Product

Definition

For vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, the 1×1 matrix $\mathbf{x}^T\mathbf{y}$ (also treated as a scalar) is called the ***scalar product*** of x and y, and is given by

$$\mathbf{x}^T\mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

- Compare with other products we have seen earlier:
  - Scalar multiplication: scalar, vector → vector
  - Linear transformation: vector → vector
  - Scalar product: vector, vector → scalar
- Other names: inner product $\langle \mathbf{x}, \mathbf{y} \rangle$, dot product $\mathbf{x} \cdot \mathbf{y}$
- Example: If $\mathbf{x} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix}$, then

$$\mathbf{x}^T\mathbf{y} = \begin{bmatrix} 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix} = 3 \cdot 4 - 2 \cdot 3 + 1 \cdot 2 = 8$$

Length, Unit Vector, Distance

Definition

For a vector $\mathbf{x}$ in $\mathbb{R}^n$, the ***Euclidean length*** (norm, magnitude) $\|\mathbf{x}\|$ of $\mathbf{x}$ is defined to be

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}} = \sqrt{{x_1}^2 + {x_2}^2 + \cdots + {x_n}^2}$$

- A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a ***unit vector*** if and only if $\|\mathbf{u}\|=1$.
- If $\mathbf{x}$ is a nonzero vector, then $\frac{\mathbf{x}}{\|\mathbf{x}\|}$ is a unit vector in the direction of $\mathbf{x}$.
- The ***distance*** between two vectors $\mathbf{x}$ and $\mathbf{y}$ is

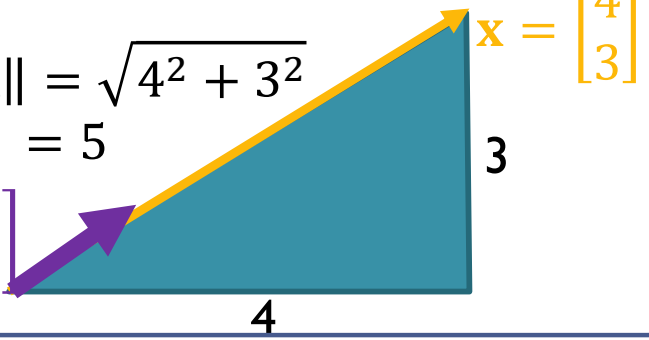
$$\|\mathbf{x}-\mathbf{y}\| = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2}$$

Example

$$\|\mathbf{x}\| = \sqrt{4^2+3^2} = 5$$

$$\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|} = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$



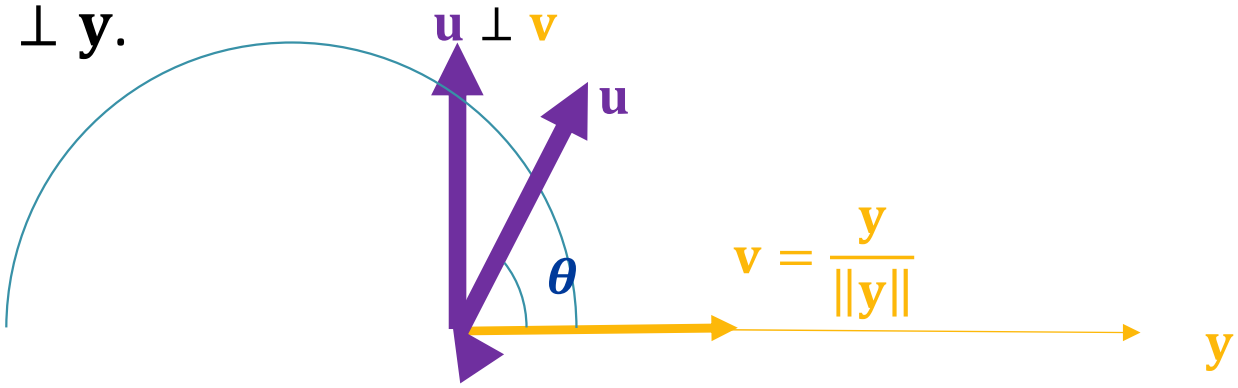
Angle and Orthogonality

Let $\mathbf{x}$ and $\mathbf{y}$ be vectors $\mathbb{R}^n$.

- Let the ***angle*** between the two vectors be θ $(0 \le \theta \le \pi)$, then $\mathbf{x}^T\mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$, i.e. if $\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}, \mathbf{v} = \frac{\mathbf{y}}{\|\mathbf{y}\|}$

$$\cos\theta = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \mathbf{u}^T\mathbf{v}$$

  - If $\cos\theta = 1$, then $\theta = 0$, $\mathbf{x}$ and $\mathbf{y}$ point in the same direction
  - If $\cos\theta = -1$, then $\theta = \pi$, $\mathbf{x}$ and $\mathbf{y}$ point in the opposite direction
  - If $\cos\theta = 0$, then $\theta = \pi/2$, $\mathbf{x}$ and $\mathbf{y}$ are perpendicular垂直/ at a right angle直角.
- If $\mathbf{x}^T\mathbf{y} = 0$, then we say that $\mathbf{x}$ and $\mathbf{y}$ are ***orthogonal正交***, and denote $\mathbf{x} \perp \mathbf{y}$.

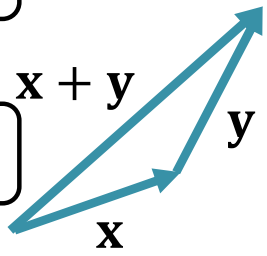


Formulas about vector lengths

Let $\mathbf{x}$ and $\mathbf{y}$ be vectors $\mathbb{R}^n$.

- $|\mathbf{x}^T\mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$. ***Cauchy-Schwarts Inequality***

Idea: $-1 \leq \cos\theta \leq 1$



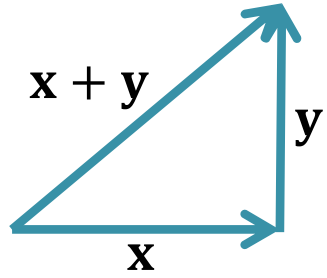
- $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. ***Triangle Inequality***

Idea: Draw a triangle

- If $\mathbf{x} \perp \mathbf{y}$, then
  $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$ ***Pythagorean Law***

Proof:

$$(\mathbf{x}+\mathbf{y})^{\mathrm{T}}(\mathbf{x}+\mathbf{y}) = \mathbf{x}^{\mathbf{T}}\mathbf{x} + \mathbf{y}^{\mathbf{T}}\mathbf{x} + \mathbf{x}^{\mathbf{T}}\mathbf{y} + \mathbf{y}^{\mathbf{T}}\mathbf{y} = \mathbf{x}^{\mathbf{T}}\mathbf{x} + \mathbf{y}^{\mathbf{T}}\mathbf{y}$$



Projecting vector onto a line

Given $\mathbf{x}$ and $\mathbf{y}$, want to write $\mathbf{x} = \mathbf{p} + \mathbf{z}$, so that $\mathbf{p}$ is parallel to $\mathbf{y}$, and $\mathbf{z}$ is orthogonal to $\mathbf{y}$.

Write $\mathbf{p} = \alpha\mathbf{u}$, where $\mathbf{u} = \mathbf{y}/\|\mathbf{y}\|$ is the unit vector in the direction of $\mathbf{y}$, and α is the length of $\mathbf{p}$. *To find α, see the picture on the right.*

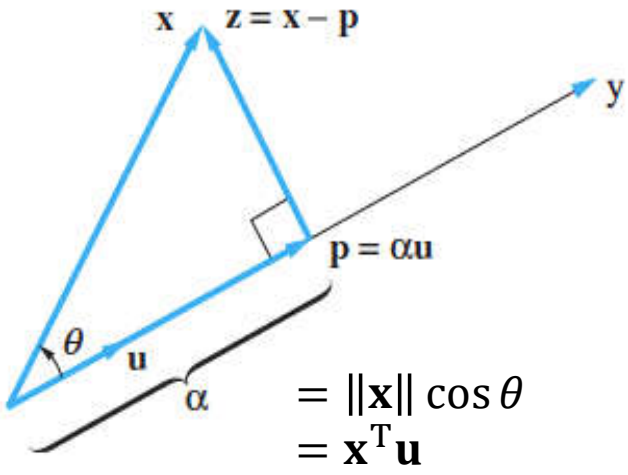
If $\mathbf{u}$ is a unit vector, then

The *scalar projection* of $\mathbf{x}$ onto $\mathbf{u}$ is

$$\alpha = \mathbf{x}^{\mathrm{T}}\mathbf{u}.$$

The ***vector projection*** of $\mathbf{x}$ onto $\mathbf{u}$ is

$$\mathbf{p} = \alpha\mathbf{u} = \left(\mathbf{x}^{\mathrm{T}}\mathbf{u}\right)\mathbf{u}.$$



The *scalar projection* of $\mathbf{x}$ onto $\mathbf{y}$ is $\alpha = \dfrac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{y}\|}$.

The ***vector projection*** of $\mathbf{x}$ onto $\mathbf{y}$ is $\mathbf{p} = \alpha\dfrac{\mathbf{y}}{\|\mathbf{y}\|} = \dfrac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}$.

Closest point and Shortest distance

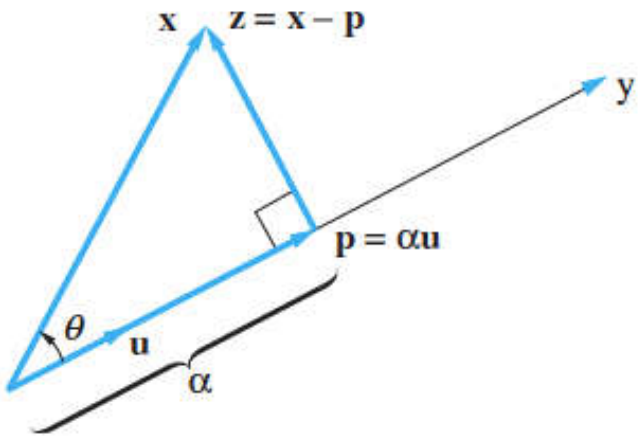
Let $\mathbf{x}$ be a point, and $l = \text{Span}(\mathbf{y})$ be a line passing through the origin.

- If $\mathbf{y}$ is any vector along the direction of line l, then the point on l closest to $\mathbf{x}$ is the ***vector projection*** of $\mathbf{x}$ onto $\mathbf{y}$:

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}$$

- If $\mathbf{w}$ is any vector orthogonal to l, then the shortest distance from point $\mathbf{x}$ to line l is the ***scalar projection*** of $\mathbf{x}$ onto $\mathbf{w}$ (with absolute value):

$$d = \frac{|\mathbf{x}^T\mathbf{w}|}{\|\mathbf{w}\|}.$$



Example: closest point 1

The point Q is the point on the line $y = \frac{1}{3}x$ that is closest to the point $\begin{bmatrix}1\\4\end{bmatrix}$. Determine the coordinates of Q.

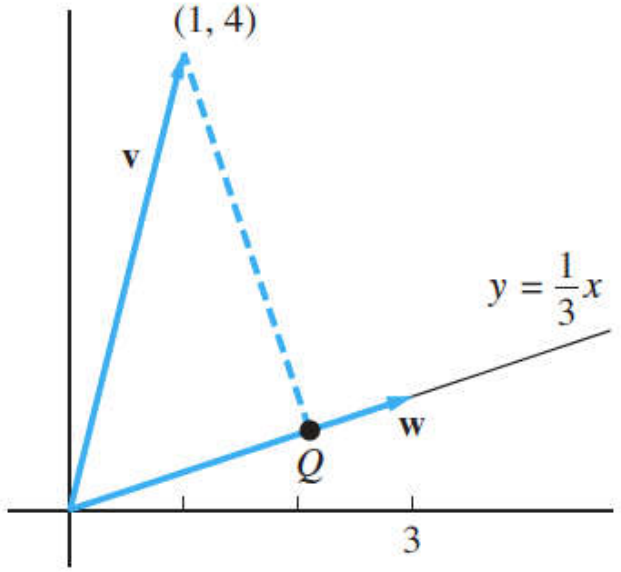
Solution

The vector $\mathbf{w} = \begin{bmatrix}3\\1\end{bmatrix}$ is a vector in the direction of line $y = \frac{1}{3}x$. Let $\mathbf{v} = \begin{bmatrix}1\\4\end{bmatrix}$, we can calculate the vector projection:

$$\frac{\mathbf{v}^T\mathbf{w}}{\mathbf{w}^T\mathbf{w}}\mathbf{w} = \frac{7}{10}\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}2.1\\0.7\end{bmatrix}$$

Since the line goes through the origin, the projection point lies on the line.

The closet point Q is $[2.1,0.7]^T$.



What if the line does not contain the origin?
See next example.

Example: closest point 2

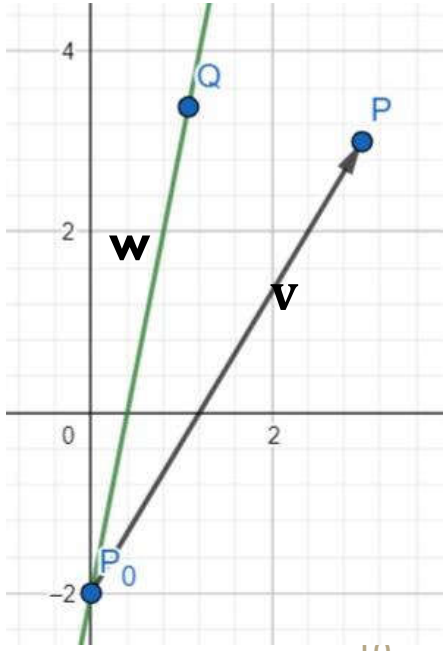
Find the point Q on the line $y = 5x - 2$ that is closest to the point $P = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$.

Solution

- Find any point P_0 on the line $y = 5x - 2$; for example, $P_0 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$.
- The vector $\mathbf{w} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ is in the direction of the line.
- $\mathbf{v} = \overrightarrow{P_0P} = P - P_0 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. Then $\overrightarrow{P_0Q}$ is the vector projection of $\mathbf{v}$ onto $\mathbf{w}$:

$$\overrightarrow{P_0Q} = \frac{\mathbf{v}^T\mathbf{w}}{\mathbf{w}^T\mathbf{w}}\mathbf{w} = \frac{3 \cdot 1 + 5 \cdot 5}{1^2 + 5^2}\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 14/13 \\ 70/13 \end{bmatrix}$$

- Therefore, the closest point is $Q = P_0 + \overrightarrow{P_0Q} = \begin{bmatrix} 14/13 \\ 44/13 \end{bmatrix}$.



Example: shortest distance

What is this shortest distance from the point $P = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ to the line $y = 5x - 2$?

Solution

- Find any point Q on the line $y = 5x - 2$; for example, Q= $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$.
- The vector $\mathbf{w} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$ is a vector in the direction that is orthogonal to the line $5x - y - 2 = 0$
- The distance is the scalar projection of $\mathbf{v} = \overrightarrow{QP} = P - Q = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ onto $\mathbf{w}$.

$$d = \frac{|\mathbf{v}^T \mathbf{w}|}{\|\mathbf{w}\|} = \frac{|3 \cdot 5 + 5 \cdot (-1)|}{\sqrt{26}} = \frac{10}{\sqrt{26}} = \frac{5}{13}\sqrt{26}$$